Speech recognition for English Language (STT&TTS) – High Accuracy AI-based Virtual Assistant

This project focuses on creating an advanced speech assistant specifically for the English language that is designed to be highly accurate in recognizing speech, correcting grammar and vocabulary, and speaking back corrected sentences fluently.

It uses HTML, CSS, and JavaScript to build a browser-based interface. The assistant utilizes the Web Speech API for speech recognition, the Speech Synthesis API for text-to-speech, and the OpenAI GPT-3.5-Turbo model to enhance grammar and vocabulary, resulting in natural and fluent output.

By combining these technologies, the assistant becomes a lightweight, deployable, and accurate tool suitable for real-time English language support directly from the browser.

Speech recognition technology has made significant strides over the past decade; however, challenges remain in achieving near-human accuracy, particularly for the English language, due to its diverse accents, pronunciations, and contextual complexities. In 2024, several approaches and advancements have emerged to improve English speech recognition accuracy.

Project Details

This project focuses on enhancing English speech recognition accuracy using modern deep learning techniques and NLP-based applications, mainly transformers and self-supervised learning. It is suitable as a team project for 2-4 members, allowing parallel work on data preparation, model fine-tuning, and testing.

Models and Methods

- Transformer Models: Whisper, wav2vec 2.0, HuBERT

- Techniques: Fine-tuning with domain-specific and accent-rich datasets, data augmentation (noise injection, speed and pitch perturbation), and noise reduction techniques.

Accuracy and Results

- Achieved Word Error Rate (WER) reductions of about 20-30% compared to earlier ASR systems.

- Better recognition of multiple accents and robust performance in noisy environments.

- Low latency for real-time applications.

Sources

- Whisper (2022), wav2vec 2.0 (2020), HuBERT (2021).

- IEEE Transactions on Audio, Speech, and Language Processing.

Requirements

- Python 3.x (recommended for deep learning projects because of large library support).

- Libraries: PyTorch, Transformers.

-GPU hardware.

- Datasets like Libri Speech and Common Voice

Recommended Language and Why

Python is recommended due to its vast ecosystem of deep learning libraries (e.g., PyTorch, TensorFlow) and strong community support, which makes experimentation and deployment easier.

Guidance for Development

1. Collect and preprocess data covering diverse accents.
2. Fine-tune a pre-trained model (e.g., Whisper) using this data.
3. Test using standard ASR metrics like WER.
4. Optimize for low latency if real-time recognition is required.
5. Document results and iterate for improvement.

References

1. Radford et al., "Whisper: Robust Speech Recognition via Large-Scale Weak Supervision" (2022)

2. Baevski et al., "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations" (2020)

3. Hsu et al., "HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units" (2021)

4. IEEE Transactions on Audio, Speech, and Language Processing (Recent Issues)

Challenges Ahead

- Handling spontaneous speech with hesitations and fillers.

- Recognizing speech in multi-speaker scenarios.

- Improving performance for underrepresented accents.

- Balancing accuracy with computational efficiency.

Conclusion

With modern transformer models, careful fine-tuning, and team collaboration, it is possible to build an English ASR system in 2024 that significantly improves accuracy and robustness over traditional approaches.